

Pregame talks

Math AA HL

Sprint 1A · Section 2.1, Lines
Two points and the whole section

Time limit: 20 minutes, strict. Start a timer. Stop when it rings, finished or not. Answers are on page 2: use them if you are stuck, and say so in the email. The point is to try.

What to send, by Tuesday 6 October, 20:00: one email to Gauti.Eggertsson@gmail.com, subject “Sprint 1”, three lines:

Done at [time]. Got to problem [n]. Looked at answers for [which, or none].

On the weekend you redo these with Gauti over Zoom, without the answers.

1. $A(1, 4)$ and $B(7, 12)$. Find the gradient of AB , the distance AB , and the midpoint of AB .
2. The line through $(1, 2)$ with gradient 3. Write its equation in the form $y = mx + c$, and then in the form $ax + by = c$ with integers.
3. The line through $P(1, 2)$ and $Q(4, 7)$, in the form $ax + by = c$ with a, b, c integers. Check your line by substituting both points.
4. (a) Are the lines $y = 3x + 5$ and $y = -\frac{1}{3}x + 8$ perpendicular? Say why. (b) Find the line through $(4, 7)$ perpendicular to $y = 3x + 5$, in the form $y = mx + c$.

Look only when stuck, and say which ones in the email. On the weekend these are redone without this page.

1. $m = \frac{12-4}{7-1} = \frac{8}{6} = \frac{4}{3}$. Distance $= \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. Midpoint $M\left(\frac{1+7}{2}, \frac{4+12}{2}\right) = (4, 8)$.

2. $y - 2 = 3(x - 1)$, so $y = 3x - 1$. Rearranged: $3x - y = 1$.

3. $m = \frac{7-2}{4-1} = \frac{5}{3}$. $y - 2 = \frac{5}{3}(x - 1)$; times 3: $3y - 6 = 5x - 5$, so $-5x + 3y = 1$ (or $5x - 3y = -1$). Check: $-5 + 6 = 1$ and $-20 + 21 = 1$.

4. (a) Yes: the gradients multiply to $3 \times (-\frac{1}{3}) = -1$, flip and change sign. (b) $m = -\frac{1}{3}$; $y - 7 = -\frac{1}{3}(x - 4)$, so $y = -\frac{1}{3}x + \frac{25}{3}$.